

PONDICHERRY UNIVERSITY
RAMANUJAN SCHOOL OF MATHEMATICAL SCIENCES
DEPARTMENT OF STATISTICS



M.Sc. QUANTITATIVE FINANCE
MSQF 532: Financial Mathematics
LECTURE NOTES ON
Financial Mathematics

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Course Objectives:

- To equip students with a solid understanding of fundamental mathematical concepts and techniques used in financial analysis, valuation of financial instruments, and portfolio management.
- To develop the ability to apply quantitative methods for solving practical problems in finance, including risk assessment, derivatives pricing, and financial decision-making.

Course Outcomes (COs):

CO1: Apply financial mathematics to evaluate investments and corporate finance decisions

CO2: Use quantitative tools for bond, equity, and portfolio valuation

CO3: Analyze risk and return to support financial strategy

CO4: Interpret and construct financial statements for cash flow management

CO5: Apply mathematical models to real-world financial problems

Unit I: Introduction to Financial Mathematics

Role and importance of mathematics in finance — Time value of money concepts — Simple interest and compound interest — Nominal and effective rates — Discounting and present value — Basic annuities: ordinary annuities, annuities due — Growth and decay curves.

Unit II: Valuation of Financial Instruments

Present value and future value of annuities — Perpetuities and growing annuities — Loan amortization and sinking funds — Bond valuation: yield to maturity (YTM), yield to call (YTC) — Equity valuation: dividend discount model (DDDM), price-earnings model.

Unit III: Financial Ratios and Cash Flow Analysis

Key financial ratios: liquidity, profitability, leverage, efficiency — Application of ratios in decision making — Cash flow statements: operating, investing, and financing cash flows — Free cash flow analysis — Break-even and sensitivity analysis.

Unit IV: Risk, Return, and Portfolio Mathematics

Measurement of return: arithmetic vs geometric mean — Standard deviation and variance of returns — Coefficient of variation — Covariance and correlation — Portfolio return and risk — Diversification and optimal portfolio construction — Capital Asset Pricing Model (CAPM).

Unit V: Advanced Topics and Applications

Linear programming in financial decision-making — Introduction to financial derivatives (options, forwards, futures) — Option valuation basics: Binomial and Black-Scholes models — Duration and convexity of bonds — Probability distributions in finance: normal, log-normal — Simulation and scenario analysis in financial planning.

Books for Reference:

1. Bodie, Z., Kane, A., & Marcus, A. J. (2022). *Investments* (12th ed.). McGraw-Hill Education.
2. Tuckman, B., & Serrat, A. (2023). *Fixed Income Securities: Tools for Today's Markets* (4th ed.). Wiley.
3. Fabozzi, F. J. (2023). *Financial Mathematics: A Comprehensive Treatment* (2nd ed.). Wiley.
4. Hull, J. C. (2024). *Options, Futures, and Other Derivatives* (11th ed.). Pearson.
5. Ross, S. A., Westerfield, R. W., Jaffe, J., & Jordan, B. D. (2022). *Corporate Finance* (13th ed.). McGraw-Hill Education.

COs	PO1	PO2	PO3	PO4	PO5
CO1	XXX	XXX	XXX	XX	XX
CO2	XXX	XXX	XXX	XX	XX
CO3	XXX	XX	XXX	XXX	XX
CO4	XX	XX	XX	XXX	XX
CO5	XXX	XXX	XXX	XX	XXX

Contents

1	Introduction to Financial Mathematics	1
1.1	Time Value of Money & Simple Interest Fundamentals	1
1.2	Theory of Compound Interest & Growth/Decay Models	4
1.3	Methods of Analysis	8
1.3.1	Compounding	8
1.3.2	Discounting & Present Value Techniques	14
1.4	Growth and Decay Curves	23

Chapter 1

Introduction to Financial Mathematics

1.1 Time Value of Money & Simple Interest Fundamentals

Lecture
13/08/2026

Definition 1.1.1: Simple Interest & Accrued Amount

Let P represent the initial **Principal** invested or borrowed, R the **nominal annual rate of interest** expressed as a percentage ($R\%$ per annum), and T the **time horizon** expressed in years.

The **Simple Interest** (SI) generated is linear in time and principal:

$$\text{SI} = \frac{P \times R \times T}{100} = P \cdot r \cdot T \quad (1.1)$$

where $r = \frac{R}{100}$ is the effective decimal interest rate per year.

The **Total Accrued Amount** (A or FV) at the end of duration T is given by:

$$A = P + \text{SI} = P(1 + rT) \quad (1.2)$$

Time Conversion Rules for Fractional Periods

When the time period is given in units other than full years:

- **Days to Years (Exact Year):** $T = \frac{\text{Number of Days}}{365}$
- **Weeks to Years:** $T = \frac{\text{Number of Weeks}}{52}$
- **Months to Years:** $T = \frac{\text{Number of Months}}{12}$

Additivity of Simple Interest

Under simple interest, the interest earned in each period depends only on the original principal P and not on previously accrued interest (i.e., interest does not compound). Consequently, over multiple disjoint consecutive time periods $[0, t_1], [t_1, t_2], \dots, [t_{k-1}, t_k]$ with respective rates $R_1, R_2, \dots, R_k$, the total simple interest is strictly additive:

$$\text{SI}_{\text{total}} = \sum_{j=1}^k \frac{P \times R_j \times \Delta t_j}{100} \quad (1.3)$$

Example 1.1.1: Basic Simple Interest and Total Amount**Problem:**

Find the simple interest and the total amount to be paid on a principal sum of Rs. 15,000 at an interest rate of 5% per annum after 2 years.

Given Data:

- Principal (P) = Rs. 15,000
- Rate (R) = 5% per annum
- Time (T) = 2 years

Step-by-Step Calculation:**1. Simple Interest (SI):**

$$SI = \frac{P \times R \times T}{100} = \frac{15,000 \times 5 \times 2}{100} = \text{Rs. } 1,500$$

2. Total Amount (A):

$$A = P + SI = 15,000 + 1,500 = \text{Rs. } 16,500$$

Final Result:

$$SI = \text{Rs. } 1,500, \quad A = \text{Rs. } 16,500$$

Example 1.1.2: Fractional Period in Days (Exact Day Count)**Problem:**

Find the simple interest and accumulated amount on a principal of Rs. 25,000 at an annual rate of 5% for a duration of 219 days.

Given Data:

- Principal (P) = Rs. 25,000
- Rate (R) = 5% p.a.
- Time (T) = 219 days = $\frac{219}{365}$ years = 0.6 years

Calculation:

$$SI = \frac{P \times R \times T}{100} = \frac{25,000 \times 5 \times \frac{219}{365}}{100} = \frac{25,000 \times 5 \times 0.6}{100} = \text{Rs. } 750$$

Total Amount:

$$A = P + SI = 25,000 + 750 = \text{Rs. } 25,750$$

Final Result:

$$SI = \text{Rs. } 750, \quad A = \text{Rs. } 25,750$$

Example 1.1.3: Fractional Period in Weeks**Problem:**

Find the simple interest and accumulated amount on a principal of Rs. 25,000 at an annual rate of 5% for a period of 20 weeks.

Given Data:

- Principal (P) = Rs. 25,000
- Rate (R) = 5% p.a.
- Time (T) = 20 weeks = $\frac{20}{52}$ years = $\frac{5}{13}$ years

Calculation:

$$SI = \frac{P \times R \times T}{100} = \frac{25,000 \times 5 \times \frac{20}{52}}{100} = \frac{1,250 \times 20}{52} \approx \text{Rs. } 480.77$$

Total Amount:

$$A = P + SI = 25,000 + 480.77 = \text{Rs. } 25,480.77$$

Final Result:

$$SI = \text{Rs. } 480.77, \quad A = \text{Rs. } 25,480.77$$

Example 1.1.4: Year-by-Year Multi-Period Breakdown**Problem:**

A principal amount of Rs. 1,00,000 is invested at a rate of 10% simple interest per annum for 3 years. Calculate the simple interest generated individually for the 1st year, 2nd year, and 3rd year, and compute the total accrued amount.

Given Data:

- Principal (P) = Rs. 1,00,000
- Rate (R) = 10% p.a.
- Total Horizon (T) = 3 years

Year-Wise Computation: Because simple interest is non-compounding, the interest generated in any single 1-year increment ($\Delta t = 1$) remains invariant:

$$\text{Year 1: } SI_1 = \frac{1,00,000 \times 10 \times 1}{100} = \text{Rs. } 10,000$$

$$\text{Year 2: } SI_2 = \frac{1,00,000 \times 10 \times 1}{100} = \text{Rs. } 10,000$$

$$\text{Year 3: } SI_3 = \frac{1,00,000 \times 10 \times 1}{100} = \text{Rs. } 10,000$$

Aggregate Interest and Terminal Amount:

$$SI_{\text{total}} = SI_1 + SI_2 + SI_3 = 10,000 + 10,000 + 10,000 = \text{Rs. } 30,000$$

$$A = P + SI_{\text{total}} = 1,00,000 + 30,000 = \text{Rs. } 1,30,000$$

Final Summary Table:

Period	Principal (Rs.)	Interest (Rs.)	Accrued Amount (Rs.)
Year 1	1,00,000	10,000	1,10,000
Year 2	1,00,000	10,000	1,20,000
Year 3	1,00,000	10,000	1,30,000
Total	—	30,000	1,30,000

Example 1.1.5: Inverse Problem: Tiered Rate Schedule

Problem:

A total simple interest of Rs. 2,500 is earned by investing an unknown sum of money (P) for a total duration of 13 years under a tiered rate structure:

- 4% p.a. for the first 3 years
- 5% p.a. for the next 4 years
- 8% p.a. for the remaining period beyond 7 years

Determine the initial principal sum invested.

Interval Breakdown: Let P denote the unknown principal sum. The total horizon of $T = 13$ years is decomposed into:

1. **Interval 1:** $T_1 = 3$ years, $R_1 = 4\%$

$$SI_1 = \frac{P \times 4 \times 3}{100} = 0.12 P$$

2. **Interval 2:** $T_2 = 4$ years, $R_2 = 5\%$

$$SI_2 = \frac{P \times 5 \times 4}{100} = 0.20 P$$

3. **Interval 3:** $T_3 = 13 - (3 + 4) = 6$ years, $R_3 = 8\%$

$$SI_3 = \frac{P \times 8 \times 6}{100} = 0.48 P$$

Linear Formulation and Algebraic Solution: The cumulative interest satisfies:

$$\begin{aligned} SI_{\text{total}} &= SI_1 + SI_2 + SI_3 \\ 2,500 &= 0.12 P + 0.20 P + 0.48 P \\ 2,500 &= 0.80 P \\ \Rightarrow P &= \frac{2,500}{0.80} = \text{Rs. } \mathbf{3,125} \end{aligned}$$

Final Result: The initial principal sum invested is Rs. **3,125**.

1.2 Theory of Compound Interest & Growth/Decay Models

Definition 1.2.1: Compound Interest & Terminal Value

Under **Compound Interest**, interest accrued at the end of each compounding period is added to the principal to form the new principal base for subsequent periods (interest on interest).

For a principal P invested at a nominal annual interest rate of $R\%$ ($r = R/100$) for n years compounded annually, the terminal accumulated amount (A or FV) is:

$$A = P \left(1 + \frac{R}{100}\right)^n = P(1 + r)^n \quad (1.4)$$

The total **Compound Interest** (CI) generated over the horizon is:

$$CI = A - P = P \left[\left(1 + \frac{R}{100}\right)^n - 1 \right] = P [(1 + r)^n - 1] \quad (1.5)$$

Exponential Growth and Decay / Depreciation Curves

- **Compound Growth / Appreciation:** If a quantity grows at constant annual percentage rate $r = R/100$:

$$A(n) = P(1 + r)^n \quad (1.6)$$

- **Compound Decay / Reducing Balance Depreciation:** If an asset or demographic metric depreciates at annual rate $d = R/100$:

$$A(n) = P(1 - d)^n \quad (1.7)$$

The **Compound Depreciation** (total value reduction) over n periods is:

$$\text{Total Depreciation} = P - A(n) = P [1 - (1 - d)^n] \quad (1.8)$$

- **Varying Multi-Year Growth/Decay Factors:** For sequential periods with rates $r_1, r_2, \dots, r_k$:

$$A_k = P \cdot (1 \pm r_1)(1 \pm r_2) \cdots (1 \pm r_k) \quad (1.9)$$

Example 1.2.1: Annual Compounding of Interest

Problem:

Find the compound interest (CI) on a principal sum of Rs. 12,600 for 2 years at an annual interest rate of 10%, compounded annually.

Given Data:

- Principal (P) = Rs. 12,600
- Rate (R) = 10% per annum
- Horizon (n) = 2 years (compounded annually)

Step-by-Step Calculation:

1. Terminal Accumulated Amount (A):

$$\begin{aligned} A &= P \left(1 + \frac{R}{100}\right)^n \\ A &= 12,600 \left(1 + \frac{10}{100}\right)^2 = 12,600(1.10)^2 \\ A &= 12,600 \times 1.21 = \text{Rs. } 15,246 \end{aligned}$$

2. Compound Interest (CI):

$$CI = A - P$$

$$CI = 15,246 - 12,600 = \text{Rs. } 2,646$$

Final Result:

$$\text{Accumulated Amount } A = \text{Rs. } 15,246, \quad CI = \text{Rs. } 2,646$$

Example 1.2.2: Compound Depreciation of an Asset (Reducing Balance)**Problem:**

A television set was purchased for Rs. 21,100. The asset depreciates in market value at a constant rate of 5% per annum. Determine the residual book value of the TV after 3 years and compute the total compound depreciation incurred.

Given Data:

- Initial Purchase Value (P) = Rs. 21,100
- Annual Rate of Depreciation (d) = 5% per annum
- Horizon (n) = 3 years

Calculation:**1. Residual Value after 3 Years (A):**

$$A = P \left(1 - \frac{d}{100} \right)^n$$

$$A = 21,100 \left(1 - \frac{5}{100} \right)^3 = 21,100(0.95)^3$$

$$A = 21,100 \times 0.857375 = \text{Rs. } 18,091.6125 \approx \text{Rs. } 18,091.61$$

2. Total Compound Depreciation:

$$\text{Total Depreciation} = P - A$$

$$= 21,100 - 18,091.61 = \text{Rs. } 3,008.39$$

Final Result:

$$\text{Residual Value after 3 Years} = \text{Rs. } 18,091.61, \quad \text{Compound Depreciation} = \text{Rs. } 3,008.39$$

Example 1.2.3: Sub-Annual Compounding: Semi-Annual Conversion**Problem:**

Find the compound interest and accumulated amount on a principal of Rs. 48,000 for 1 year at an annual nominal rate of 8%, compounded half-yearly.

Given Data:

- Principal (P) = Rs. 48,000

- Nominal Annual Rate (R) = 8% p.a.
- Horizon (n) = 1 year
- Compounding Frequency (m) = 2 periods per year (half-yearly)

Parameter Conversions:

- Periodic Interest Rate: $i_{\text{periodic}} = \frac{R}{m} = \frac{8\%}{2} = 4\% = 0.04$
- Total Compounding Cycles: $N = n \times m = 1 \times 2 = 2$ periods

Calculation:**1. Accumulated Terminal Amount (A):**

$$A = P \left(1 + \frac{R/m}{100} \right)^{n \cdot m}$$

$$A = 48,000 \left(1 + \frac{4}{100} \right)^2 = 48,000(1.04)^2$$

$$A = 48,000 \times 1.0816 = \text{Rs. } 51,916.80$$

2. Compound Interest (CI):

$$CI = A - P$$

$$CI = 51,916.80 - 48,000 = \text{Rs. } 3,916.80$$

Final Result:

Terminal Amount $A = \text{Rs. } 51,916.80$, $CI = \text{Rs. } 3,916.80$

Example 1.2.4: Demographic Modeling: Inverse Tiered Population Decay**Problem:**

The population of a town decreases every year due to migration, economic shifts, and unemployment. The present population is 6,31,680. The rate of migration was 4% during the past year and 6% during the year preceding that. Determine what the population was 2 years ago.

Given Data:

- Current Population (A_2) = 6,31,680
- Rate of reduction in Year 1 (two years ago, d_1) = 6%
- Rate of reduction in Year 2 (last year, d_2) = 4%
- Let the initial population 2 years ago = P

Algebraic Formulation & Solution: Applying sequential geometric reduction factors:

$$P \times \left(1 - \frac{d_1}{100} \right) \times \left(1 - \frac{d_2}{100} \right) = A_2$$

$$P \times \left(1 - \frac{6}{100} \right) \times \left(1 - \frac{4}{100} \right) = 6,31,680$$

$$P \times 0.94 \times 0.96 = 6,31,680$$

$$P \times 0.9024 = 6,31,680$$

Solving for initial population P :

$$P = \frac{6,31,680}{0.9024} = 7,00,000$$

Final Result: The town population 2 years ago was **7,00,000**.

1.3 Methods of Analysis

There are two primary methods of financial time-value analysis:

Lecture
17/08/2026

1. **Compounding**
2. **Discounting**

1.3.1 Compounding

Compounding over Single and Multiple Conversion Periods

a. Compounding of Interest over n Years

$$\text{Future Value (FV)} = P(1 + R)^n \quad (1.10)$$

where:

- P = Principal amount
- R = Annual rate of interest (in decimal $r = R/100$)
- n = Number of years

b. Multiple Compounding Periods

$$\text{Future Value (FV)} = P \left(1 + \frac{R}{m} \right)^{n \cdot m} \quad (1.11)$$

where:

- P = Principal amount
- R = Nominal annual rate of interest
- n = Number of years
- m = Frequency of compounding per annum
- $n \cdot m$ = Total number of conversion periods
- $\frac{R}{m}$ = Periodic interest rate per conversion period

Lecture
24/08/2026

Example 1.3.1: Multi-Frequency Compounding Comparison

Problem: Calculate the maturity amount if Rs. 2,00,000 is invested for 2 years at a nominal interest rate of 12% per annum compounded: (a) Annually (b) Semi-annually (c) Quarterly (d) Monthly

Given Data:

- Principal (P) = Rs. 2,00,000
- Nominal Rate (r) = 12% p.a. = 0.12
- Time Horizon (t) = 2 years

General Compounding Formula:

$$A = P \left(1 + \frac{r}{m} \right)^{m \cdot t}$$

Detailed Step-by-Step Calculations:

1. Compounded Annually ($m = 1$):

$$\begin{aligned} A &= 2,00,000 \times \left(1 + \frac{0.12}{1} \right)^{1 \times 2} = 2,00,000 \times (1.12)^2 \\ &= 2,00,000 \times 1.2544 = \text{Rs. } \mathbf{2,50,880.00} \end{aligned}$$

2. Compounded Semi-Annually ($m = 2$):

$$\begin{aligned} A &= 2,00,000 \times \left(1 + \frac{0.12}{2} \right)^{2 \times 2} = 2,00,000 \times (1.06)^4 \\ &= 2,00,000 \times 1.26247696 = \text{Rs. } \mathbf{2,52,495.39} \end{aligned}$$

3. Compounded Quarterly ($m = 4$):

$$\begin{aligned} A &= 2,00,000 \times \left(1 + \frac{0.12}{4} \right)^{4 \times 2} = 2,00,000 \times (1.03)^8 \\ &= 2,00,000 \times 1.26677008 = \text{Rs. } \mathbf{2,53,354.02} \end{aligned}$$

4. Compounded Monthly ($m = 12$):

$$\begin{aligned} A &= 2,00,000 \times \left(1 + \frac{0.12}{12} \right)^{12 \times 2} = 2,00,000 \times (1.01)^{24} \\ &= 2,00,000 \times 1.26973465 = \text{Rs. } \mathbf{2,53,946.93} \end{aligned}$$

Summary Comparison Table:

Frequency	Periods (m)	Periodic Rate (r/m)	Cycles ($m \cdot t$)	Maturity Amount (A)
Annually	1	12.00%	2	Rs. 2,50,880.00
Semi-annually	2	6.00%	4	Rs. 2,52,495.39
Quarterly	4	3.00%	8	Rs. 2,53,354.02
Monthly	12	1.00%	24	Rs. 2,53,946.93

Effective Rate of Interest (ERI)

c. Effective Rate of Interest (ERI) Annually compounded rate of interest that is equivalent to an annual interest rate compounded more than once per year:

$$\text{ERI} = \left(1 + \frac{R}{m} \right)^m - 1 \quad (1.12)$$

where:

- ERI = Effective rate of interest
- R = Nominal rate of interest
- m = Frequency of compounding per year

Doubling Period Formulations

d. Doubling Period

[Lecture: 18/08/2026]

The **Doubling Period** is the time required for an investment to double its principal amount at a given rate of interest.

- **Continuous Compounding Basis:** Based on the natural logarithm of 2:

$$\ln(2) \approx 0.693 \quad (69.3\%) \quad (1.13)$$

- **Rule of 72 (Standard Practical Thumb Rule):**

$$\text{Doubling Period} = \frac{72}{\text{Rate of Interest } (R)} \quad (1.14)$$

$$\text{Rate of Interest } (R) = \frac{72}{\text{Doubling Period}} \quad (1.15)$$

- **Rule of 69 (Refined Discrete Compounding Approximation):**

$$\text{Doubling Period} = 0.35 + \frac{69}{R} \quad (1.16)$$

Lecture
25/08/2026

Example 1.3.2: Determination of Interest Rate using Rule of 72

Problem: Calculate the rate of interest if the doubling period of an investment is: (a) 4 years (b) 6 years

Formula:

$$\text{Rate of Interest } (R) = \frac{72}{\text{Doubling Period}}$$

Calculations:

1. For a doubling period of 4 years:

$$R = \frac{72}{4} = 18\% \text{ p.a.}$$

2. For a doubling period of 6 years:

$$R = \frac{72}{6} = 12\% \text{ p.a.}$$

Lecture
25/08/2026

Example 1.3.3: Determination of Doubling Period using Rule of 69

Problem: Calculate the doubling period using the Rule of 69 if the annual rate of interest is: (a) 6% (b) 9% (c) 12%

Formula:

$$\text{Doubling Period} = 0.35 + \frac{69}{R}$$

Calculations:

1. If interest rate is $R = 6\%$:

$$\text{Doubling Period} = 0.35 + \frac{69}{6} = 0.35 + 11.50 = \mathbf{11.85} \text{ years}$$

2. If interest rate is $R = 9\%$:

$$\text{Doubling Period} = 0.35 + \frac{69}{9} = 0.35 + 7.667 = \mathbf{8.02} \text{ years}$$

3. If interest rate is $R = 12\%$:

$$\text{Doubling Period} = 0.35 + \frac{69}{12} = 0.35 + 5.75 = \mathbf{6.10} \text{ years}$$

Compound Value of a Series of Payments

e. Compound Value of a Series of Payments For an uneven stream of payments P_t invested across horizon n :

$$\text{Compound Value (FV)} = \sum_{t=1}^n P_t(1 + R)^{n-t} \quad (1.17)$$

Lecture
25/08/2026

Example 1.3.4: Compound Value of an Uneven Series of Payments

Problem: Calculate the Future Value (FV) of the following series of payments made at the end of each year over a 4-year investment horizon at an interest rate of $R = 9\%$ per annum:

- $P_1 = \text{Rs. } 1,500$ at the end of the 1st year
- $P_2 = \text{Rs. } 3,000$ at the end of the 2nd year
- $P_3 = \text{Rs. } 4,500$ at the end of the 3rd year
- $P_4 = \text{Rs. } 6,000$ at the end of the 4th year

Analytical Formulation: Payments deposited at the end of year t earn compound interest for $(n - t)$ remaining years until the terminal horizon $n = 4$:

$$\text{FV} = \sum_{t=1}^4 P_t(1 + R)^{4-t} = P_1(1 + R)^3 + P_2(1 + R)^2 + P_3(1 + R)^1 + P_4(1 + R)^0$$

Step-by-Step Payment Accumulation:

- **1st Payment (Rs. 1,500):** Invested for $4 - 1 = 3$ years:

$$FV_1 = 1,500 \times (1 + 0.09)^3 = 1,500 \times 1.295029 = \text{Rs. } \mathbf{1,942.55}$$

- **2nd Payment (Rs. 3,000):** Invested for $4 - 2 = 2$ years:

$$FV_2 = 3,000 \times (1 + 0.09)^2 = 3,000 \times 1.188100 = \text{Rs. } \mathbf{3,564.30}$$

- **3rd Payment (Rs. 4,500):** Invested for $4 - 3 = 1$ year:

$$FV_3 = 4,500 \times (1 + 0.09)^1 = 4,500 \times 1.090000 = \text{Rs. } \mathbf{4,905.00}$$

- **4th Payment (Rs. 6,000):** Invested for $4 - 4 = 0$ years (at terminal date):

$$FV_4 = 6,000 \times (1 + 0.09)^0 = 6,000 \times 1.000000 = \text{Rs. } \mathbf{6,000.00}$$

Compound Value Factor (CVAF) Summary Table:

Year (t)	Payment (P_t)	Horizon ($4 - t$)	Factor $(1.09)^{4-t}$	Compound Value (FV_t)
1	Rs. 1,500	3 yrs	1.295029	Rs. 1,942.55
2	Rs. 3,000	2 yrs	1.188100	Rs. 3,564.30
3	Rs. 4,500	1 yr	1.090000	Rs. 4,905.00
4	Rs. 6,000	0 yrs	1.000000	Rs. 6,000.00
Total Terminal Future Value (FV):				Rs. 16,411.85

Final Result: The aggregate compound value of the series of payments at the end of 4 years is **Rs. 16,411.85**.

Compound Value of an Annuity

f. Compound Value of an Annuity For a sequence of equal constant periodic cash flows A invested at interest rate R over duration n :

$$FV \text{ of Annuity} = A \times \left[\frac{(1 + R)^n - 1}{R} \right] = A \cdot CVAF(R, n) \quad (1.18)$$

where:

- A = Constant periodic cash flow or annuity amount
- R = Periodic interest rate
- n = Total number of compounding annuity conversion periods
- $CVAF(R, n)$ = Compound Value Annuity Factor = $\frac{(1 + R)^n - 1}{R}$

Lecture
27/08/2026

Example 1.3.5: Compound Value of an Annual Annuity

Problem: Calculate the maturity value (Future Value) of an annuity if Rs. 20,000 is paid annually for 7 years at an interest rate of 12% per annum compounded annually.

Given Data:

- Periodic Annuity Flow (A) = Rs. 20,000
- Duration (n) = 7 years
- Interest Rate (R) = 12% p.a. = 0.12

Analytical Solution:

$$FV = A \times \left[\frac{(1 + R)^n - 1}{R} \right] = 20,000 \times \left[\frac{(1 + 0.12)^7 - 1}{0.12} \right]$$

Evaluating the compounding power factor $(1.12)^7 \approx 2.2106814$:

$$\begin{aligned} FV &= 20,000 \times \left[\frac{2.2106814 - 1}{0.12} \right] = 20,000 \times \left[\frac{1.2106814}{0.12} \right] \\ &= 20,000 \times 10.089012 = \text{Rs. } \mathbf{2,01,780.24} \end{aligned}$$

Table Approximation Note: Using standard 3-decimal compound annuity factor tables (CVAF(12%, 7) $\approx$ 10.089 or rounded value 10.090):

$$FV \approx 20,000 \times 10.090 = \text{Rs. } \mathbf{2,01,800.00}$$

Final Result: The maturity accumulated value of the annuity after 7 years is Rs. **2,01,780.24** (approx. Rs. **2,01,800.00**).

Lecture
27/08/2026

Example 1.3.6: Maturity Value of Monthly Recurring Deposit Annuity

Problem: Bhuvanesh pays a monthly installment of Rs. 2,000 towards a recurring deposit with a financier. The rate of interest is 6% per annum compounded monthly. What is the maturity value of this recurring deposit after 10 months?

Given Data:

- Monthly Deposit (A) = Rs. 2,000
- Nominal Annual Rate (R) = 6% p.a.
- Periodic Monthly Rate (i) = $\frac{6\%}{12} = 0.5\% = 0.005$
- Total Number of Monthly Instalments (n) = 10 months

Analytical Solution:

$$FV = A \times \left[\frac{(1 + i)^n - 1}{i} \right] = 2,000 \times \left[\frac{(1 + 0.005)^{10} - 1}{0.005} \right]$$

Evaluating $(1.005)^{10} \approx 1.05114013$:

$$\begin{aligned} FV &= 2,000 \times \left[\frac{1.05114013 - 1}{0.005} \right] = 2,000 \times \left[\frac{0.05114013}{0.005} \right] \\ &= 2,000 \times 10.228026 = \text{Rs. } \mathbf{20,456.05} \end{aligned}$$

Approximation Note: Using standard recurring deposit financial tables (CVAF(0.5%, 10) $\approx$ 10.22):

$$FV \approx 2,000 \times 10.22 = \text{Rs. } \mathbf{20,440.00}$$

Final Result: The maturity value of the recurring deposit after 10 months is Rs. **20,456.05** (approx. Rs. **20,440.00**).

1.3.2 Discounting & Present Value Techniques

Present Value of Lump Sum & Series of Cash Flows

Lecture
18/08/2026

a. Present Value of a Lump Sum The present value (PV) of a single future cash flow FV discounted across n periods at rate R :

$$PV = FV \times \frac{1}{(1 + R)^n} = FV \cdot (1 + R)^{-n} = FV \cdot PVF(R, n) \quad (1.19)$$

b. Present Value of a Series of Cash Flows For an uneven sequence of future cash inflows CF_t received across periods $t = 1, 2, \dots, n$:

$$PV = \sum_{t=1}^n \frac{CF_t}{(1 + R)^t} = \sum_{t=1}^n CF_t \times (1 + R)^{-t} \quad (1.20)$$

Lecture
27/08/2026

Example 1.3.7: Present Value of a Single Future Lump Sum

Problem: Find the present value of Rs. 10,000 receivable after 4 years at an interest rate of 12% per annum. (The PV factor in the standard table for 12% and 4 years is 0.6355).

Given Data:

- Future Value (FV) = Rs. 10,000
- Duration (n) = 4 years
- Discount Rate (R) = 12% p.a. = 0.12

Calculation:

$$PV = FV \times \frac{1}{(1 + R)^n} = 10,000 \times \frac{1}{(1 + 0.12)^4} = 10,000 \times \frac{1}{1.573519}$$

$$PV = 10,000 \times 0.635518 = \text{Rs. } \mathbf{6,355.18}$$

Using 3-Decimal Table Factor (0.635):

$$PV = 10,000 \times 0.635 = \text{Rs. } \mathbf{6,350.00}$$

Final Result: The present value of Rs. 10,000 receivable after 4 years is Rs. **6,355.18** (or Rs. **6,350.00** using table factor).

Lecture
27/08/2026

Example 1.3.8: Construction of Present Value Factor Schedule (12% p.a.)

Problem: Calculate the Present Value Factors (PVF) at an interest rate of 12% per annum for a period of 5 years.

Formula:

$$PVF(12\%, n) = \frac{1}{(1 + 0.12)^n} = (1.12)^{-n}$$

Schedule of Discount Factors:

Year (n)	Mathematical Computation	Present Value Factor (PVF)
1	$\frac{1}{(1.12)^1} = \frac{1}{1.120000}$	0.8929
2	$\frac{1}{(1.12)^2} = \frac{1}{1.254400}$	0.7972
3	$\frac{1}{(1.12)^3} = \frac{1}{1.404928}$	0.7118
4	$\frac{1}{(1.12)^4} = \frac{1}{1.573519}$	0.6355
5	$\frac{1}{(1.12)^5} = \frac{1}{1.762342}$	0.5674

Present Value of Annuities (Ordinary Annuity vs. Annuity Due)**c. Present Value of an Annuity (Ordinary Annuity – Period-End Payments)**

$$PV \text{ of Annuity} = \text{Annuity Amount } (A) \times \left[\frac{1 - (1 + R)^{-n}}{R} \right] = A \times PVAF(R, n) \quad (1.21)$$

d. Present Value of Annuity Due (Period-Beginning Payments)

$$PV \text{ of Annuity Due} = \text{Annuity Amount } (A) \times PVAF(R, n) \times (1 + R) \quad (1.22)$$

Lecture
27/08/2026**Example 1.3.9: Present Value of an Ordinary Annuity Stream**

Problem: Raja has entered into an agreement that will fetch him Rs. 60,000 per annum for the next 4 years. Calculate the present value of these future cash inflows at a discount rate of 20% per annum.

Given Data:

- Annual Cash Inflow (A) = Rs. 60,000
- Horizon (n) = 4 years
- Discount Rate (R) = 20% p.a. = 0.20

Method 1: Year-by-Year Cash Flow Discounting Table

Year (t)	Cash Inflow (CF_t)	PV Factor at 20%: $\frac{1}{(1.20)^t}$	Present Value (PV_t)
1	Rs. 60,000	$\frac{1}{(1.20)^1} = 0.833333$	Rs. 50,000.00
2	Rs. 60,000	$\frac{1}{(1.20)^2} = 0.694444$	Rs. 41,666.67
3	Rs. 60,000	$\frac{1}{(1.20)^3} = 0.578704$	Rs. 34,722.22
4	Rs. 60,000	$\frac{1}{(1.20)^4} = 0.482253$	Rs. 28,935.19
Total Aggregate Present Value (PV):			Rs. 1,55,324.08

Method 2: Analytical PVAF Closed-Form Formula

$$\begin{aligned}
 PV &= A \times \left[\frac{1 - (1 + R)^{-n}}{R} \right] = 60,000 \times \left[\frac{1 - (1.20)^{-4}}{0.20} \right] \\
 &= 60,000 \times \left[\frac{1 - 0.482253}{0.20} \right] = 60,000 \times \left[\frac{0.517747}{0.20} \right] \\
 &= 60,000 \times 2.588735 = \text{Rs. } 1,55,324.08
 \end{aligned}$$

Table Factor Approximation Note: Using standard Present Value Annuity Table factor for 4 years at 20% ($PVAF(20\%, 4) \approx 2.589$):

$$PV \approx 60,000 \times 2.589 = \text{Rs. } 1,55,340.00$$

Final Result: The present value of Raja's annuity agreement is Rs. 1,55,324.08 (or Rs. 1,55,340.00 using 3-decimal table factor).

Lecture
28/08/2026

Example 1.3.10: Present Value of an Annuity Due

Problem: Mr. Thanay expects to receive Rs. 20,000 at the **beginning** of each year for 4 years. Calculate the present value of this annuity due assuming an annual interest rate of 9%.

Given Data:

- Annual Instalment (A) = Rs. 20,000
- Horizon (n) = 4 payments made at $t = 0, 1, 2, 3$
- Discount Rate (R) = 9% p.a. = 0.09

Method 1: Year-by-Year Discounting of Beginning-of-Year Receipts Because payments occur at the beginning of each year, the first payment is received at $t = 0$ (no discounting):

- **Beginning of Year 1 ($t = 0$):** $PV_0 = \text{Rs. } 20,000$
- **Beginning of Year 2 ($t = 1$):** $PV_1 = \frac{20,000}{(1.09)^1} = \text{Rs. } 18,348.62$
- **Beginning of Year 3 ($t = 2$):** $PV_2 = \frac{20,000}{(1.09)^2} = \text{Rs. } 16,833.60$

- **Beginning of Year 4 ($t = 3$):** $PV_3 = \frac{20,000}{(1.09)^3} = \text{Rs. } 15,443.67$

Summing all individual present values:

$$PV_{\text{due}} = 20,000.00 + 18,348.62 + 16,833.60 + 15,443.67 = \text{Rs. } \mathbf{70,625.89}$$

Method 2: Annuity Due Analytical Formula

$$\begin{aligned} PV_{\text{due}} &= A \times \left[\frac{1 - (1 + R)^{-n}}{R} \right] \times (1 + R) \\ &= 20,000 \times \left[\frac{1 - (1.09)^{-4}}{0.09} \right] \times 1.09 \\ &= 20,000 \times \left[\frac{1 - 0.708425}{0.09} \right] \times 1.09 \\ &= 20,000 \times 3.239720 \times 1.09 = \text{Rs. } \mathbf{70,625.89} \approx \text{Rs. } \mathbf{70,626.00} \end{aligned}$$

Final Result: The present value of Mr. Thanay's annuity due is Rs. **70,625.89** (approx. Rs. **70,626.00**).

Present Value of a Constant Perpetuity

e. Present Value of a Perpetuity (Constant Perpetuity – Infinite Horizon) A **Perpetuity** is an annuity whose periodic cash flows continue indefinitely. The present value of a constant perpetual stream of cash flows C discounted at rate R per period is:

$$PV \text{ of Constant Perpetuity} = \frac{C}{R} \quad (1.23)$$

where:

- C = Constant periodic cash inflow (e.g. annual coupon, preferred dividend)
- R = Discount / capitalization rate per payment period

Lecture
31/08/2026

Example 1.3.11: Valuation of Preference Shares as a Constant Perpetuity

Problem: A company has issued 10% preference shares with a face value of Rs. 100 each. What would be the fair market price (Present Value) of the preference share if the required rate of return for the investor is 8% per annum?

Given Data:

- Face Value of Preference Share = Rs. 100
- Annual Preference Dividend Rate = 10%
- Constant Annual Dividend (C) = 10% $\times$ Rs. 100 = Rs. 10.00
- Required Rate of Return / Discount Rate (R) = 8% = 0.08

Analytical Solution: Since preference shares pay a constant periodic dividend indefinitely without a specified maturity date, their value is given by the constant perpetuity formula:

$$PV = \frac{C}{R} = \frac{10}{0.08} = \text{Rs. } \mathbf{125.00}$$

Conclusion: The investor will be willing to pay up to Rs. **125.00** to purchase the preference share to earn a 10% dividend on its face value at their required 8% yield.

Lecture
31/08/2026

Example 1.3.12: Present Value of a Perpetual Cash Flow Stream

Problem: Let us assume that an investor expects a perpetual annual sum of Rs. 500 from an investment. What is the present value of this perpetuity if the benchmark interest rate is 10% per annum?

Given Data:

- Perpetual Annual Cash Inflow (C) = Rs. 500
- Discount Rate (R) = 10% = 0.10

Analytical Calculation:

$$PV = \frac{C}{R} = \frac{500}{0.10} = \text{Rs. } 5,000.00$$

Final Result: The present value of the perpetual Rs. 500 annual cash stream at 10% discount rate is Rs. **5,000.00**.

Present Value of a Growing Perpetuity

f. Present Value of a Growing Perpetuity A **Growing Perpetuity** is an infinite stream of periodic cash flows that grow at a constant compound rate G per period. When the discount rate exceeds the growth rate ($R > G$):

$$PV \text{ of Growing Perpetuity} = \frac{C_1}{R - G} \quad (1.24)$$

where:

- C_1 = Expected cash inflow at the end of the first period
- R = Discount rate / required rate of return per period
- G = Constant compound growth rate of cash flows per period
- **Condition for Convergence:** $R > G$ (if $R \leq G$, the present value diverges to infinity).

Lecture
31/08/2026

Example 1.3.13: Valuation of a Growing Dividend Stream (Gordon Growth Perpetuity)

Problem: A company is expected to declare a dividend of Rs. 4.00 at the end of the 1st year from now (C_1), and this dividend is projected to grow at a constant annual rate of 5% indefinitely. What is the present value of this perpetual dividend stream if the investor's required rate of return is 10% per annum?

Given Data:

- First-Year Dividend (C_1) = Rs. 4.00

- Constant Annual Dividend Growth Rate (G) = 5% = 0.05
- Required Rate of Return / Discount Rate (R) = 10% = 0.10

Convergence Check: Since $R = 0.10 > G = 0.05$, the finite valuation condition ($R > G$) is satisfied.

Analytical Solution: Applying the growing perpetuity (Gordon growth) formula:

$$\begin{aligned} PV &= \frac{C_1}{R - G} = \frac{4}{0.10 - 0.05} \\ &= \frac{4}{0.05} = \text{Rs. } \mathbf{80.00} \end{aligned}$$

Final Result: The present value (fair market price) of the perpetual growing dividend stream is Rs. **80.00**.

Present Value of a Growing Annuity

g. Present Value of a Growing Annuity (Finite Horizon n) For a stream of n periodic cash flows growing at rate G per period across a finite horizon (with $R \neq G$):

$$PV \text{ of Growing Annuity} = \frac{C_1}{R - G} \left[1 - \left(\frac{1 + G}{1 + R} \right)^n \right] \quad (1.25)$$

where:

- C_1 = Cash flow at period 1
- R = Discount rate per period
- G = Growth rate per period
- n = Life / duration of the annuity

Lecture
28/08/2026

Example 1.3.14: Present Value of a Growing Annuity Stream

Problem: Mr. Deva has rented out a commercial shop for 4 years at an initial annual rent of Rs. 12,000. The tenant has agreed to an escalation clause where the rent increases by 10% every year. If the required rate of return (discount rate) is 8%, find the present value of the expected series of rental payments.

Given Data:

- Initial First-Year Rent (C_1) = Rs. 12,000
- Lease Duration (n) = 4 years
- Annual Rent Escalation Rate (G) = 10% = 0.10
- Discount Rate (R) = 8% = 0.08

Method 1: Year-by-Year Escalated Cash Flow Discounting Table

- **Year 1:** Cash Flow C_1 = Rs. 12,000

$$PV_1 = \frac{12,000}{(1.08)^1} = \frac{12,000}{1.08} = \text{Rs. } \mathbf{11,111.11}$$

- **Year 2:** Cash Flow $C_2 = 12,000 \times (1 + 0.10) = \text{Rs. } 13,200$

$$PV_2 = \frac{13,200}{(1.08)^2} = \frac{13,200}{1.1664} = \text{Rs. } \mathbf{11,316.87}$$

- **Year 3:** Cash Flow $C_3 = 13,200 \times (1 + 0.10) = \text{Rs. } 14,520$

$$PV_3 = \frac{14,520}{(1.08)^3} = \frac{14,520}{1.259712} = \text{Rs. } 11,526.54$$

- **Year 4:** Cash Flow $C_4 = 14,520 \times (1 + 0.10) = \text{Rs. } 15,972$

$$PV_4 = \frac{15,972}{(1.08)^4} = \frac{15,972}{1.360489} = \text{Rs. } 11,740.16$$

Summing all discounted cash inflows:

$$PV = 11,111.11 + 11,316.87 + 11,526.54 + 11,740.16 = \text{Rs. } 45,694.68$$

Method 2: Growing Annuity Analytical Closed-Form Formula When growth rate $G > R$:

$$\begin{aligned} PV &= \frac{C_1}{G - R} \left[\left(\frac{1 + G}{1 + R} \right)^n - 1 \right] = \frac{12,000}{0.10 - 0.08} \left[\left(\frac{1.10}{1.08} \right)^4 - 1 \right] \\ &= \frac{12,000}{0.02} \left[(1.0185185)^4 - 1 \right] = 6,00,000 \times [1.0761578 - 1] \\ &= 6,00,000 \times 0.0761578 = \text{Rs. } 45,694.68 \end{aligned}$$

Final Result: The present value of the 4-year escalating rental stream is **Rs. 45,694.68**.

Sinking Fund (S.F.) Accumulation & Periodic Instalment

h. Sinking Fund (S.F.) Accumulation A **Sinking Fund** is a dedicated reserve established by investing equal periodic installments at compound interest to accumulate a targeted future capital sum (e.g. to redeem debentures or replace capital equipment):

$$\text{Accumulated Target Fund (FV)} = \text{S.F. Instalment (A)} \times \left[\frac{(1 + R)^n - 1}{R} \right] = A \cdot \text{CVAF}(R, n) \quad (1.26)$$

The required periodic installment A is given by:

$$\text{S.F. Instalment (A)} = \frac{\text{Target Future Fund} \times R}{(1 + R)^n - 1} = \frac{\text{Target Future Fund}}{\text{CVAF}(R, n)} \quad (1.27)$$

Lecture
31/08/2026

Example 1.3.15: Determination of Annual Sinking Fund Instalment

Problem: An amount of Rs. 2,00,000 is required at the end of 5 years from now to repay a debenture liability. What amount should be accumulated (invested) at the end of each year at an annual interest rate of 12% so that the fund reaches Rs. 2,00,000 after 5 years?

Given Data:

- Target Future Liability (FV) = Rs. 2,00,000
- Accumulation Horizon (n) = 5 years
- Annual Rate of Interest (R) = 12% p.a. = 0.12

Analytical Solution: Let A denote the required annual sinking fund installment deposited at the end of each year:

$$FV = A \times \left[\frac{(1+R)^n - 1}{R} \right] \Rightarrow A = \frac{FV \times R}{(1+R)^n - 1}$$

Evaluating the compound annuity accumulation factor $CVAF(12\%, 5)$:

$$(1.12)^5 \approx 1.76234168$$

$$CVAF(12\%, 5) = \frac{1.76234168 - 1}{0.12} = \frac{0.76234168}{0.12} \approx 6.352847$$

Solving for the annual sinking fund installment A :

$$A = \frac{2,00,000}{6.352847} = \text{Rs. } \mathbf{31,481.95}$$

Table Approximation Notes:

- Using standard 3-decimal Compound Value Annuity Factor tables ($CVAF(12\%, 5) \approx 6.353$):

$$A \approx \frac{2,00,000}{6.353} = \text{Rs. } \mathbf{31,481.19}$$

- Using standard Sinking Fund Factor tables ($SFF(12\%, 5) = \frac{1}{CVAF} \approx 0.15741$):

$$A = 2,00,000 \times 0.15741 = \text{Rs. } \mathbf{31,482.00}$$

Final Result: The required annual sinking fund installment to be invested each year is Rs. **31,481.95** (approx. Rs. **31,482.00**).

Lecture
01/09/2026

Example 1.3.16: Annuity Due Sinking Fund (T Limited)

Problem: T Limited is creating a sinking fund to redeem its debentures of Rs. 40,00,000 issued on April 5, 2008 and maturing on April 4, 2020. The first annual payment to the fund will be made on April 5, 2008 (at the beginning of each operating year). The company will make equal annual payments and expects that the fund will earn an interest rate of 14% per annum. How much will be the amount of the annual sinking fund payment?

Given Data:

- Target Sinking Fund Liability (FV) = Rs. 40,00,000
- Debenture Issue Date: April 5, 2008 Maturity Date: April 4, 2020
- Horizon (n) = 2020 – 2008 = 12 years
- Timing: Beginning of each year (Annuity Due)
- Annual Interest Rate (R) = 14% p.a. = 0.14

Formula:

$$FV = \text{Instalment } (A) \times \left[\frac{(1+R)^n - 1}{R} \right] \times (1+R)$$

Substitution & Calculation:

$$40,00,000 = A \times \left[\frac{(1 + 0.14)^{12} - 1}{0.14} \right] \times (1 + 0.14)$$

$$40,00,000 = A \times \left[\frac{4.817906 - 1}{0.14} \right] \times 1.14$$

$$40,00,000 = A \times \left[\frac{3.817906}{0.14} \right] \times 1.14$$

$$40,00,000 = A \times 27.270756 \times 1.14$$

$$40,00,000 = A \times 31.088662$$

$$A = \frac{40,00,000}{31.088662} = \text{Rs. } \mathbf{1,28,664.31}$$

Final Result: The required annual sinking fund instalment is Rs. **1,28,664.31**.

1.4 Growth and Decay Curves

Definition 1.4.1: Exponential Growth and Decay Models

Let t denote time ($t \geq 0$) and $n(t)$ denote the modeled quantity at time t . The dynamics of continuous exponential growth and decay are characterized by:

- **Exponential Growth Model:** $n(t) = n_0 e^{kt}$, $k > 0$ (initial quantity $n_0 = n(0)$, growth constant k)
- **Exponential Decay Model:** $n(t) = n_0 e^{-kt}$, $k > 0$ (initial quantity $n_0 = n(0)$, decay constant k)

For scaling over time interval $[0, T]$ between boundary values $[n_{\min}, n_{\max}]$:

$$k = \frac{1}{T} \ln \left(\frac{n_{\max}}{n_{\min}} \right) = \frac{1}{5} \ln \left(\frac{70}{10} \right) = \frac{\ln 7}{5} \approx 0.389182 \quad (1.28)$$

- **Growth Curve:** $n(t) = 10 e^{0.389182t}$, increasing monotonically from 10 to 70.
- **Decay Curve:** $n(t) = 70 e^{-0.389182t}$, decreasing monotonically from 70 to 10.

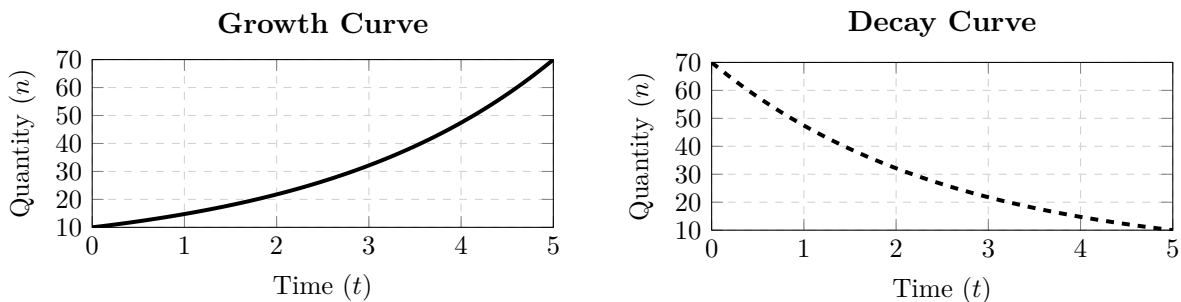


Figure 1.1: Research-Style Greyscale Exponential Growth (Solid) and Decay (Dashed) Curves

Decay Curves in Financial Trading: Price Impact Analysis

Market Microstructure Interpretation of the Decay Curve: In financial markets and quantitative execution, a **decay curve** models how a trade's impact on asset price evolves over time:

- **X-axis:** Time elapsed following trade execution.
- **Y-axis:** Price change / price impact.

Institutional Users (Large Banks): Decay curves are systematically employed by large investment banks and market-making desks to:

1. **Monitor trade impacts:** Measuring how individual block orders or algorithmic parent orders move the bid-ask quote and midpoint over time.
2. **Detect toxic behavior:** Identifying informed order flow, adverse selection, and toxic flow that degrades fill quality and market liquidity.

Core Purpose:

1. **Understand long-term trade impact:** Disentangling temporary market impact (liquidity consumption that decays back to fundamental value) from permanent price impact (information revelation).
2. **Ensure market stability:** Managing institutional execution schedules (e.g. TWAP/VWAP/POV) to prevent excessive volatility and preserve market stability.